

Supplementary Proof: Moments of the Exponential Family

This note proves that, for a distribution in the exponential family, $\mathbb{E}[Y] = b'(\theta)$ and $\text{Var}[Y] = \phi b''(\theta)$. These results are used in Chapter 4 of the notes.

Setup

A random variable Y belongs to the *exponential family* if its density (or mass function) can be written

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}, \quad (1)$$

where θ is the *canonical parameter*, $\phi > 0$ is the *dispersion parameter*, $b(\theta)$ is a twice-differentiable function, and $c(y, \phi)$ depends on y and ϕ but not on θ . We treat ϕ as known throughout.

The key tool is *differentiation under the integral sign* (Leibniz rule): for a density f that integrates to 1 we may differentiate the identity $\int f \, dy = 1$ with respect to the parameter θ .

Proof that $\mathbb{E}[Y] = b'(\theta)$

Since $f(y; \theta, \phi)$ is a valid density (or mass function), we have

$$\int f(y; \theta, \phi) \, dy = 1.$$

Differentiating both sides with respect to θ ,

$$\int \frac{\partial}{\partial \theta} f(y; \theta, \phi) \, dy = 0.$$

Using the *score identity* $\partial f / \partial \theta = f \cdot \partial \log f / \partial \theta$, this becomes

$$\int \frac{\partial \log f}{\partial \theta} f \, dy = 0, \quad \text{i.e.} \quad \mathbb{E} \left[\frac{\partial \log f}{\partial \theta} \right] = 0.$$

For the exponential family (1),

$$\frac{\partial \log f}{\partial \theta} = \frac{y - b'(\theta)}{\phi}.$$

Hence

$$\mathbb{E} \left[\frac{Y - b'(\theta)}{\phi} \right] = 0 \implies \mathbb{E}[Y] = b'(\theta). \quad \square$$

Proof that $\text{Var}[Y] = \phi b''(\theta)$

Differentiating $\mathbb{E}[\partial \log f / \partial \theta] = 0$ again with respect to θ ,

$$\mathbb{E} \left[\frac{\partial^2 \log f}{\partial \theta^2} \right] + \mathbb{E} \left[\left(\frac{\partial \log f}{\partial \theta} \right)^2 \right] = 0.$$

This identity follows from the fact that $\partial^2/\partial\theta^2 \int f dy = 0$ and that the cross-term produces the sum of the two expectations shown. The second term on the left is the *Fisher information*: $\mathcal{I}(\theta) = \mathbb{E}[(\partial \log f / \partial \theta)^2]$.

For the exponential family,

$$\frac{\partial^2 \log f}{\partial \theta^2} = -\frac{b''(\theta)}{\phi},$$

so the identity gives

$$-\frac{b''(\theta)}{\phi} + \mathcal{I}(\theta) = 0 \implies \mathcal{I}(\theta) = \frac{b''(\theta)}{\phi}.$$

On the other hand,

$$\mathcal{I}(\theta) = \mathbb{E}\left[\left(\frac{Y - b'(\theta)}{\phi}\right)^2\right] = \frac{\text{Var}[Y]}{\phi^2}.$$

Combining,

$$\frac{\text{Var}[Y]}{\phi^2} = \frac{b''(\theta)}{\phi} \implies \text{Var}[Y] = \phi b''(\theta). \quad \square$$

Remark 1. The function $V(\mu) = b''(\theta)$, expressed as a function of the mean $\mu = b'(\theta)$, is called the *variance function*. For the Normal distribution $b(\theta) = \theta^2/2$ so $V(\mu) = 1$; for the Poisson $b(\theta) = e^\theta$ so $V(\mu) = \mu$; for the Binomial with index m , $b(\theta) = m \log(1 + e^\theta)$ so $V(\mu) = \mu(1 - \mu/m)/m$.